

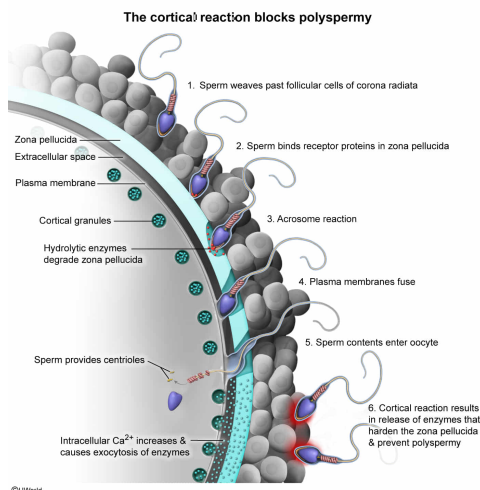
Once the sperm reaches the zona pellucida during fertilization, the cortical reaction is triggered. This aids in fertilization by:

- ✓ ☒ A. preventing more than one sperm from entering the oocyte. [62%]
☐ B. making the zona pellucida more permeable. [13%]
☐ C. activating the acrosome reaction. [12%]
☐ D. triggering the sperm to fuse with the oocyte. [11%]

Correct

62% answered correctly

Explanation:



In mammals, **fertilization** occurs when the **sperm** reaches the **oocyte**. The steps of fertilization are as follows:

1. **Sperm reaches oocyte:** The sperm weaves past follicular cells of the corona radiata.
2. **Sperm contact:** The sperm binds receptors in the **zona pellucida**, a thick matrix of glycoproteins that surrounds the oocyte.
3. **Acrosome reaction:** The **acrosome** is a specialized vesicle filled with hydrolytic enzymes, located in the sperm head. When these enzymes are released, the zona pellucida is degraded, enabling the sperm to reach the oocyte's **plasma membrane**.
4. **Fusion:** The plasma membranes of the oocyte and sperm are fused.
5. **Sperm contents enter oocyte:** The nucleus, mitochondria, and a pair of centrioles enter the oocyte.
6. **Cortical reaction:** The contents of the **cortical granules** in the oocyte are released into the space between the plasma membrane and the zona pellucida, fusing with the plasma membrane. This hardens the zona pellucida into a protective envelope, blocking additional sperm from entering.

Without the cortical reaction, more than one sperm would be able to fuse with the oocyte, a condition known as **polyspermy**. If polyspermy did occur, the resulting zygote would not have a diploid ($2n$) genome. In addition, sperm also donate centrioles, which help form the zygote's **mitotic spindle**. In polyspermy, extra mitotic spindles are formed, which can lead to aberrant segregation of **chromosomes** and embryonic death. Therefore, the cortical reaction aids in fertilization by **preventing more than one sperm from entering the oocyte**.

(Choice B) The cortical reaction acts to harden the zona pellucida, making it less (*not* more) permeable to prevent additional sperm from entering the oocyte.

(Choice C) The acrosome reaction occurs *before* the cortical reaction. Therefore, the cortical reaction does *not* activate the acrosome reaction.

(Choice D) The purpose of the cortical reaction is to create a barrier *after* the first sperm fuses to the oocyte so that additional sperm are not allowed to enter. The cortical reaction does *not* have a role in the actual fusion of the oocyte and sperm plasma membranes.

Educational objective:

During mammalian fertilization, a particular series of events leads to the fusion of the oocyte with a single sperm. When the sperm contacts the oocyte, the acrosomal reaction degrades the zona pellucida. Subsequently, in the cortical reaction, the release of cortical granules hardens the zona pellucida, preventing additional sperm from entering.

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